MSC ARTIFICIAL INTELLIGENCE DISSERTATION

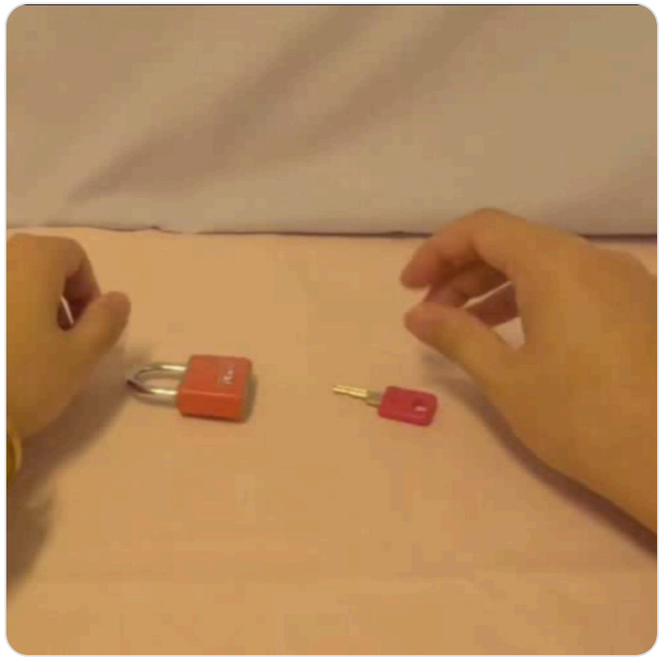
Learning to Select Actions for Complex Hand Tasks

A comparative study of visuomotor policies and latent world models

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Supervised by Prof. Alexiei Dingli
Department of Artificial Intelligence, Faculty of ICT, University of Malta

What is a policy?

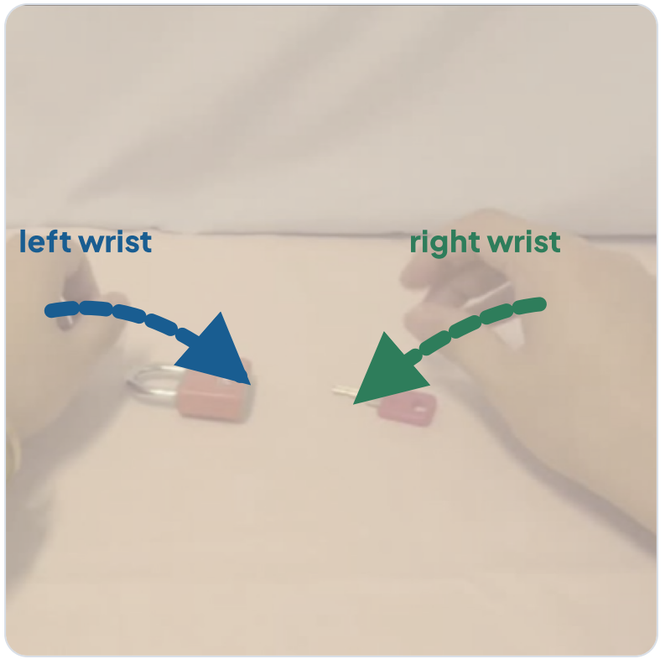
A policy is a model that learns from demonstrations (human, robot or simulated) to turn the current scene into a chunk of future actions.



In this study, input: the current scene

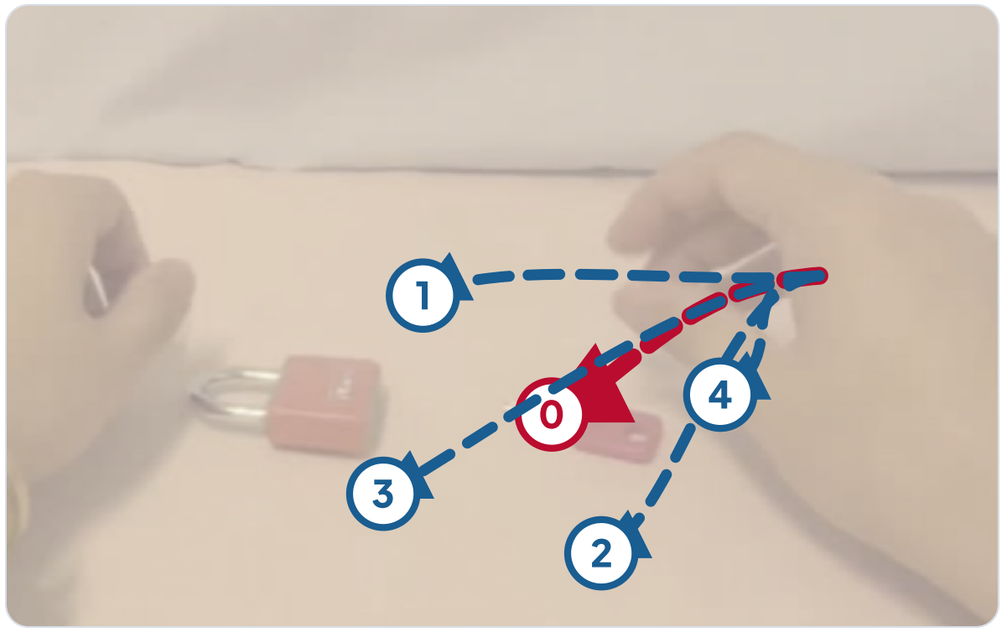
Policy

ACT or Diffusion Policy

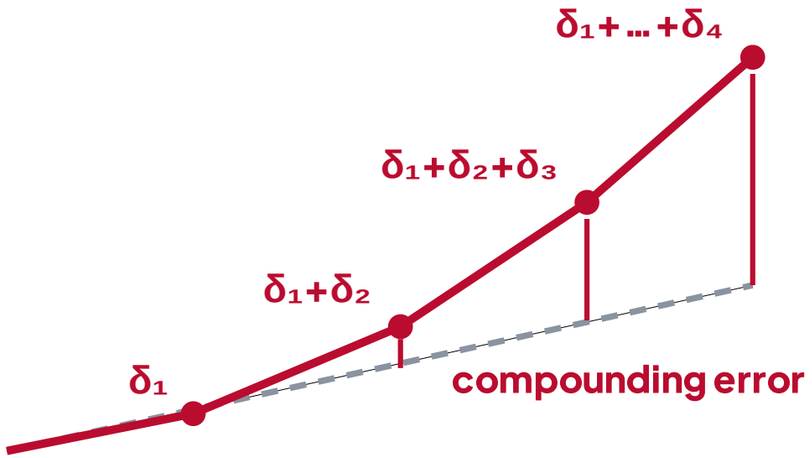


In this study, output: an action chunk, the next 16 wrist poses for both hands (18 numbers per step)

The problem of selection in a policy



Different latent or noise samples give different candidates.
Candidate 0 is the policy's default output: z = 0 for ACT, the first sample for Diffusion Policy.



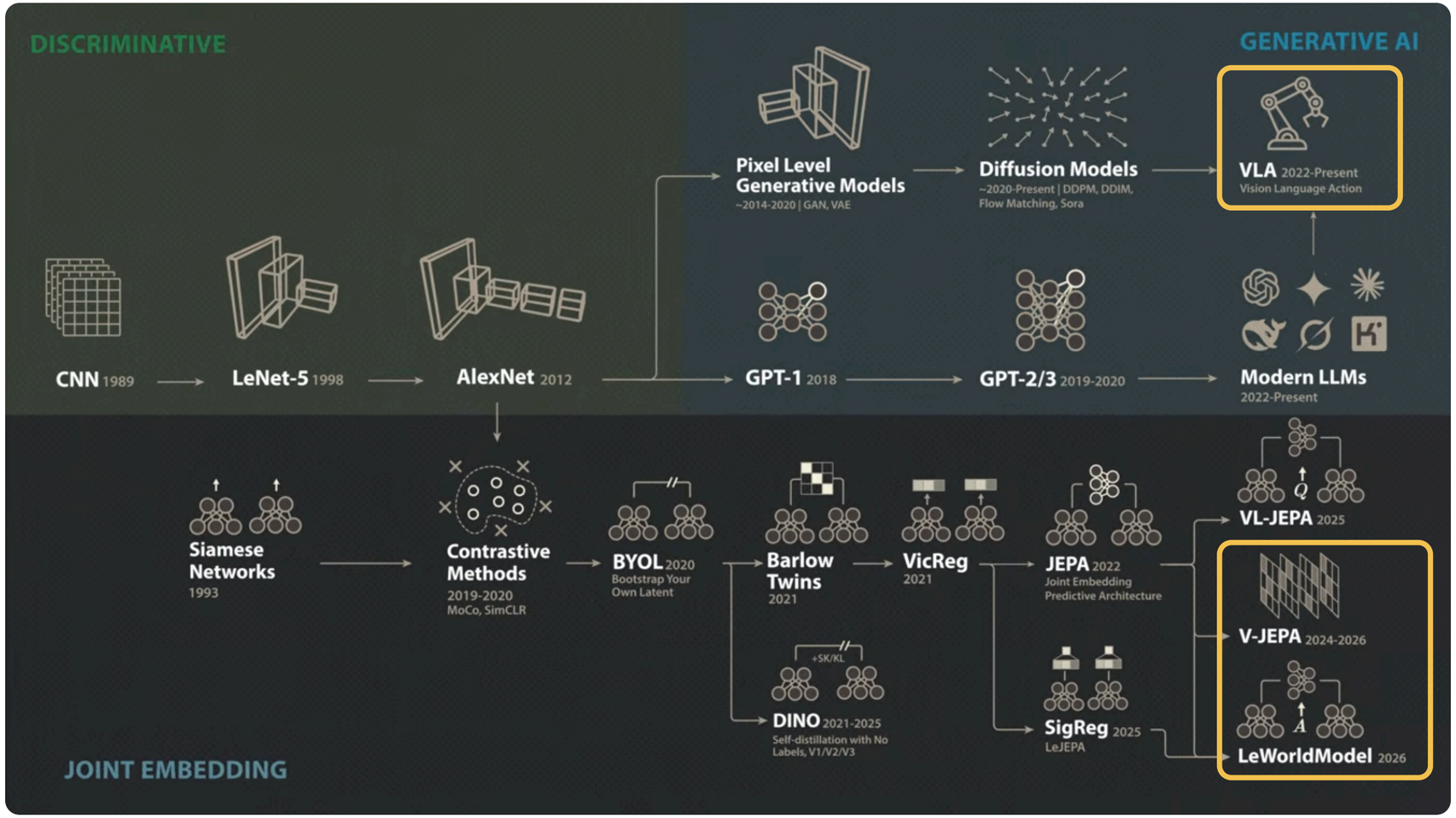
$$e_t = e_{t-1} + \delta_t = \delta_1 + \ldots + \delta_t$$

Suboptimal candidate → compounding error → unsuccessful or dangerous state

TECHNOLOGY OVERVIEW

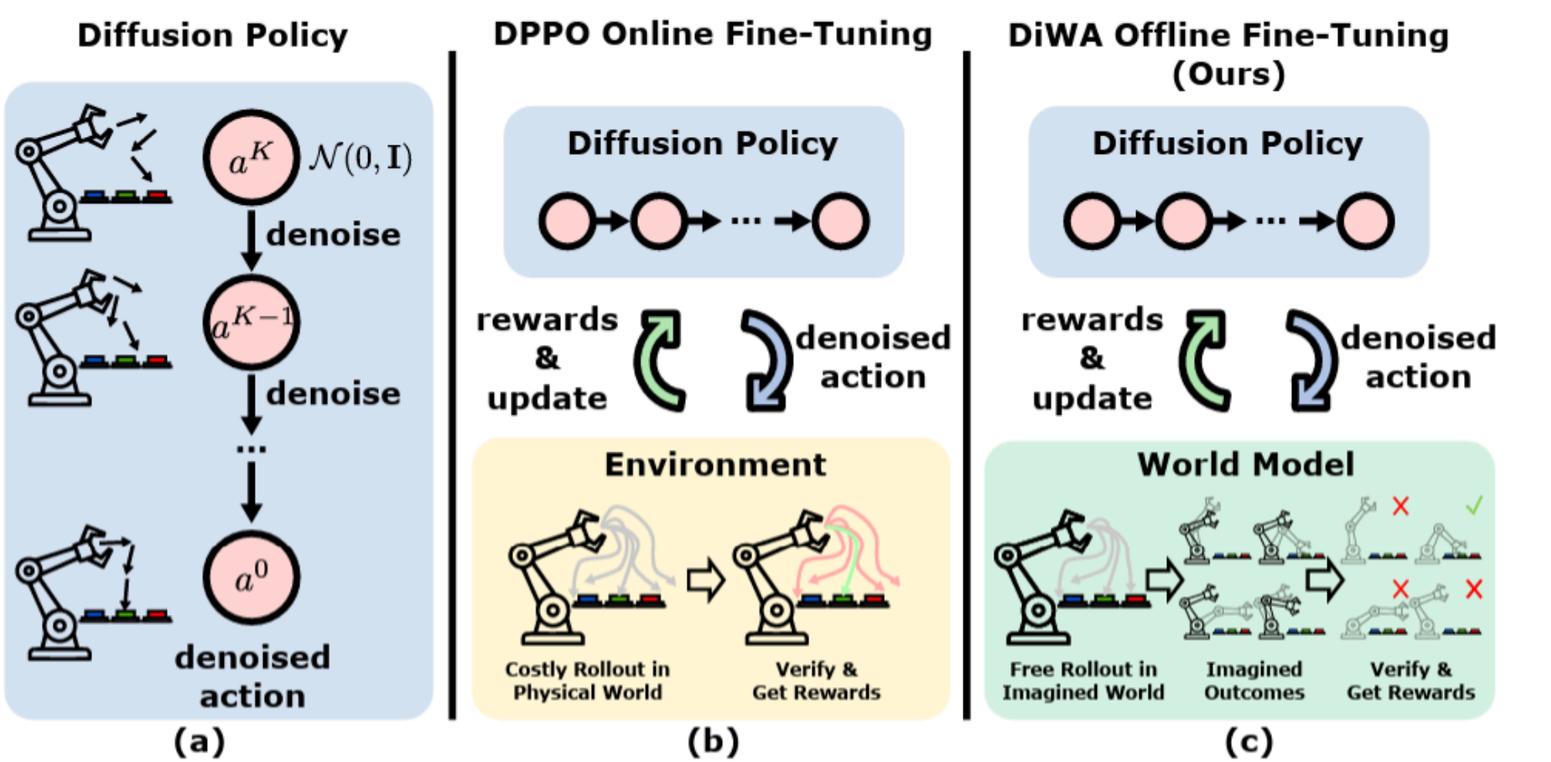
Why world models, not a VLA

A VLA maps the scene straight to an action, without predicting its consequences. A world model predicts those consequences first.



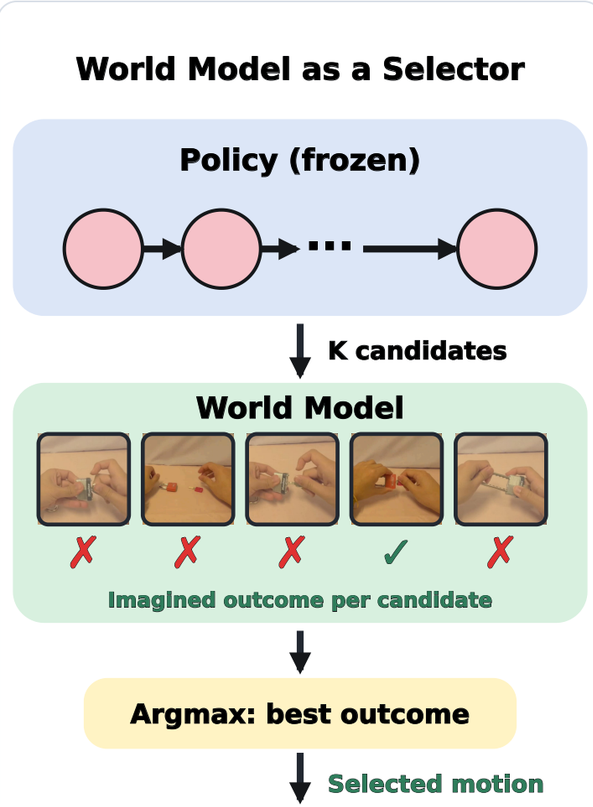
Screenshot from Welch Labs, Yann LeCun's $1B Bet Against LLMs (2026). Highlights added.

DIWA: THE PAPER WE BUILD ON



Left: Chandra et al. (2025), evaluated on CALVIN. This study reproduces (c). Right: illustrative frames.

OUR APPROACH: EXTERNAL WORLD-MODEL SELECTION



RESEARCH QUESTION

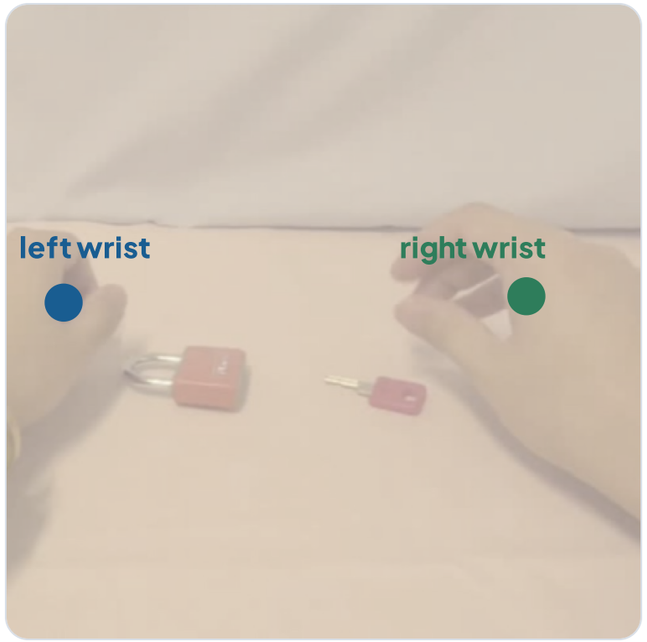
Can a frozen world model, used externally, select better motions from a policy's own candidates?

Tested offline, with the world model acting only as the final decision stage over candidates the policy already proposed.

- How much does it help, and for which policy and world-model pairings?
- When does selection fail, and why?
- How does selecting compare with finetuning the policy inside a world model (DiWA on CALVIN)?

THE DATASET

EgoDex: first-person video of two-handed tasks



Recorded with Apple Vision Pro, both hands tracked at 30 frames per second

EgoDex full release

829 hours, 338k episodes, 194 tasks

Training parts

1	2	3	4	5

Our training: parts 1 to 3

63 tasks, 194,333 episodes, 430,962 windows

not used

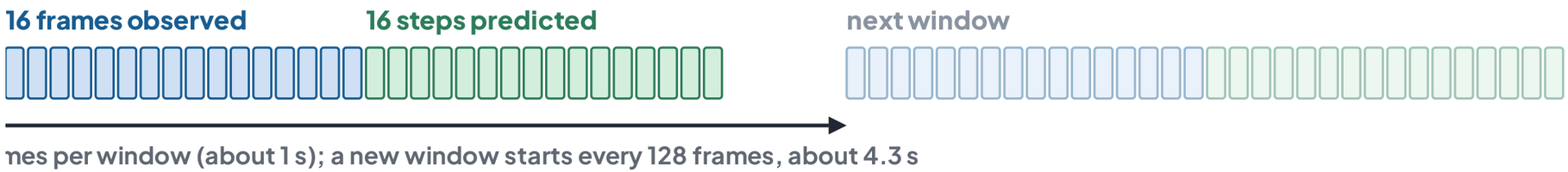
Official test set

111 tasks
3,229 episodes
7,607 windows

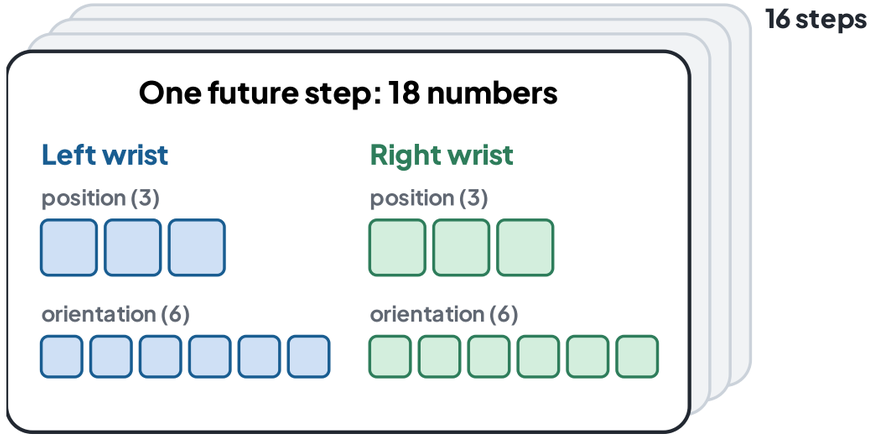
never used in training

OUR DATA

Windows of wrist motion, not video frames



nes per window (about 1 s); a new window starts every 128 frames, about 4.3 s



18 numbers per step

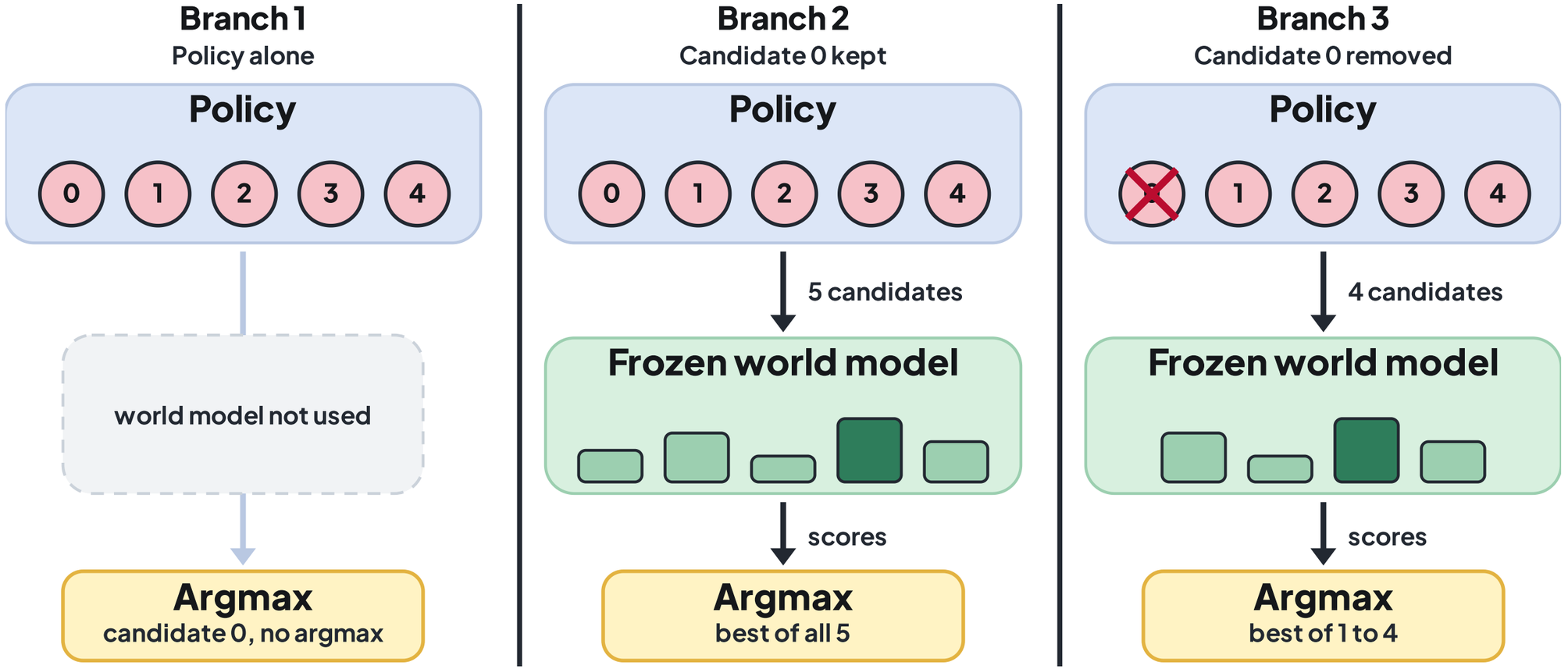
Per wrist: 3 position + 6 orientation (6-D rotation encoding)

12 DoF in total, 6 per wrist. Poses come from the headset's hand tracking.

430,962
TRAINING WINDOWS

7,607
TEST WINDOWS

THE METHOD



K = 5
CANDIDATES PER WINDOW

3 seeds
POOLED RESULTS

COMPUTE CONSTRAINTS

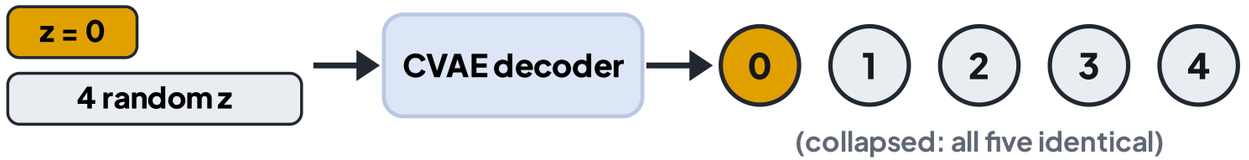
Understanding candidate selection

random, not tunable

Diffusion Policy
how it selects

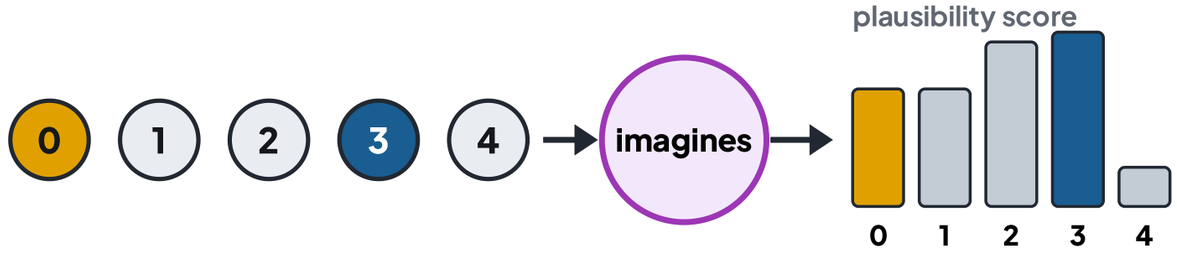


ACT
how it selects



(collapsed: all five identical)

World model
how it improves it



No selection

Candidate 0 = the policy run once.
The first sample or z = 0,
and it is kept.

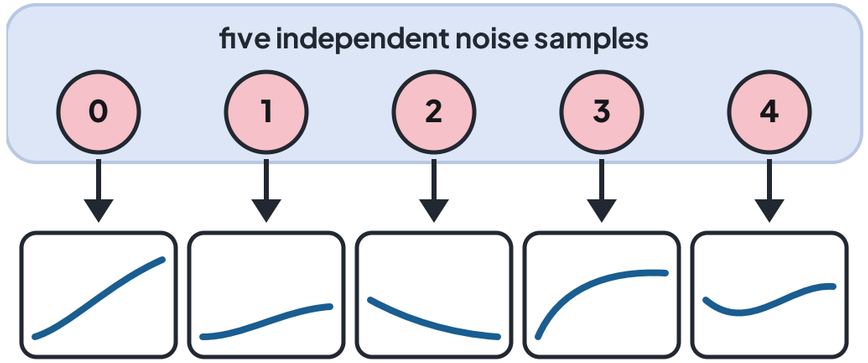
Selection

Candidate 3 has the highest plausibility score.

Never sees the real motion.
True error is measured afterwards.

POSTERIOR COLLAPSE

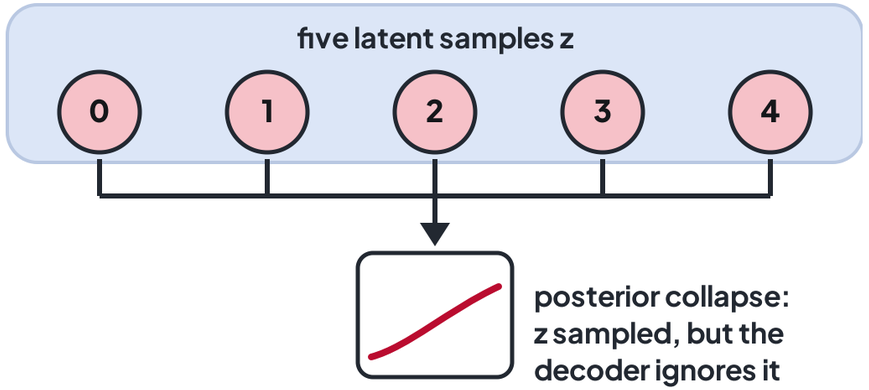
Diffusion Policy



Five genuinely different motions

independent pool, clean measurement

ACT



Five copies of one motion

no diversity, so selection has nothing to choose between

Diffusion Policy denoises Gaussian noise into a motion over DDIM steps. 5 independent noise samples give 5 candidates.

ACT is a CVAE, where a latent z sampled at test time should vary the motion. Posterior collapse means the decoder learned to ignore z, so all 5 samples are the same motion.

ACT: four latent training configurations

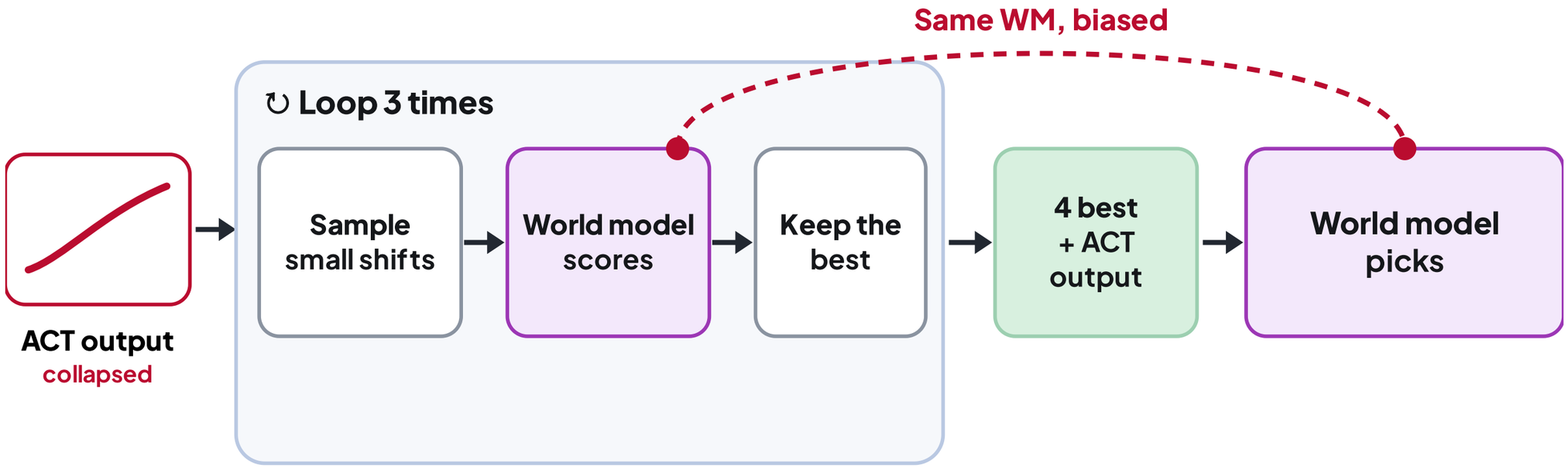
RUN	RECIPE	WHAT IT IS MEANT TO DO
v1	KL weight 10, the standard recipe	Baseline: the decoder can ignore z
v2	KL weight 100	Penalise information in z ten times harder
v3	Cyclical KL annealing and 30% observation dropout	Ease the penalty in cycles and hide observations, so the decoder must lean on z
v4	Cyclical KL annealing and free bits, 0.5 nats per dimension	Let z carry a minimum amount of information for free

No diversity

- All five samples identical
- Differences below 10^{-6}
- Posterior collapse

CEMP: Cross-Entropy Method with Policy

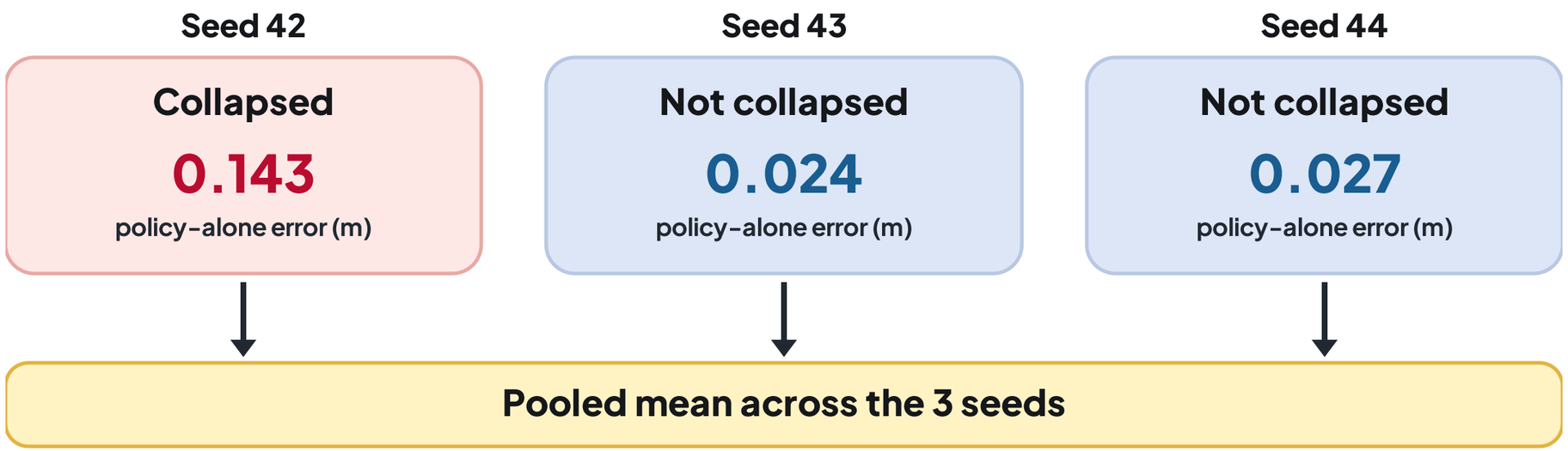
Cross-entropy method, started from the policy's output (inspired by TD-MPC2).



The same world model builds the pool and then picks from it, so the pool already leans towards what that world model prefers.

ACT results therefore measure the world model and the planner as one unit.

CEMP only applied where needed



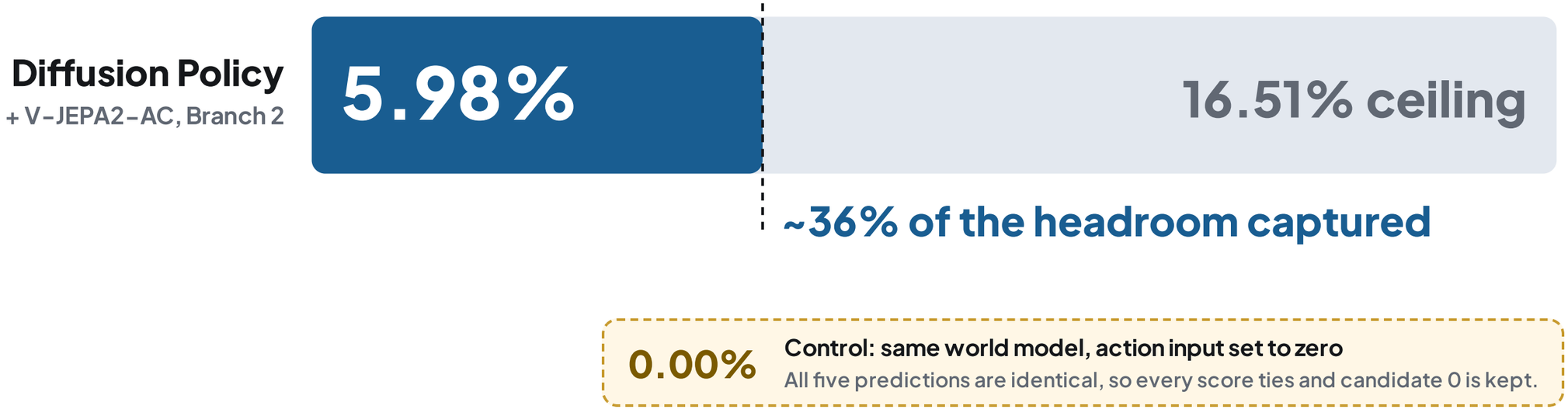
Only seed 42 needed CEMP, so seeds 43 and 44 were scored on ACT's own candidates. The pooled ACT result therefore mixes a world-model-plus-planner measurement with a plain scorer measurement. Read it as one result for this rule, not as a clean measurement of the scorer. This is why Diffusion Policy carries the main claim.

5 world models, 1 validated gain

MODEL	WHAT IT IS	GAIN	WHAT IT SHOWED
V-JEPA2-AC	Frozen video-pretrained encoder, our action head	-5.98%	Validated: its score tracks the true error
DexWM	Frozen DINOv2 encoder, second predictor and a hand-consistency head	+0.57%	No gain, single seed
LeWM	Small, compact, trained end to end, no pretrained encoder	-10.1%	Big gain, but its scores did not track the true error, so the gain is not from its predictions
V-JEPA2, no action	Control: same model, action input set to zero	0.00%	Every score ties: the motion is what matters
PointWorld proxy	Weakened motion input	+0.27%	No gain
DINO-WM, JEPA-WM	Official robot checkpoints, smoke-tested only	latent only	Interface mismatch: 18-D hand motion squeezed into 7-D robot actions

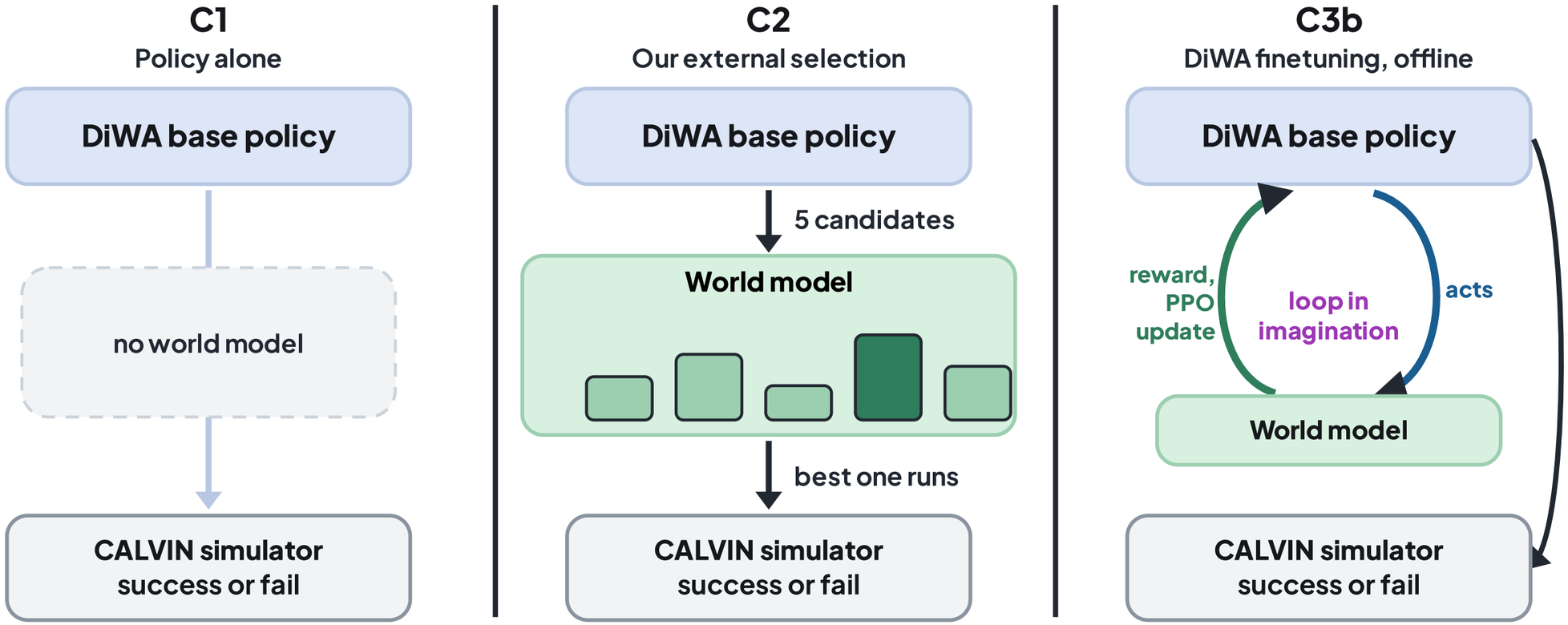
Translation error change, Diffusion Policy, Branch 2, matched training exposure. ACT with CEMP gave -0.58%, read as one unit (slides 12 to 14). LeWM detail: Backup C.

Ranking Diffusion Policy candidates with V-JEPA2-AC cuts translation error by 5.98%



The gain comes from reading each candidate's motion, not from the video backbone. Pooled across three seeds, 95% CI [5.43, 6.47], no retraining. Rotation error -0.96°. Win / tie / loss 49.9 / 23.2 / 27.0%. Branch 3: 5.37% of a 14.68% ceiling. The ceiling is what a perfect selector could remove from the same five candidates.

SECONDARY STUDY ON DIWA'S OWN BENCHMARK



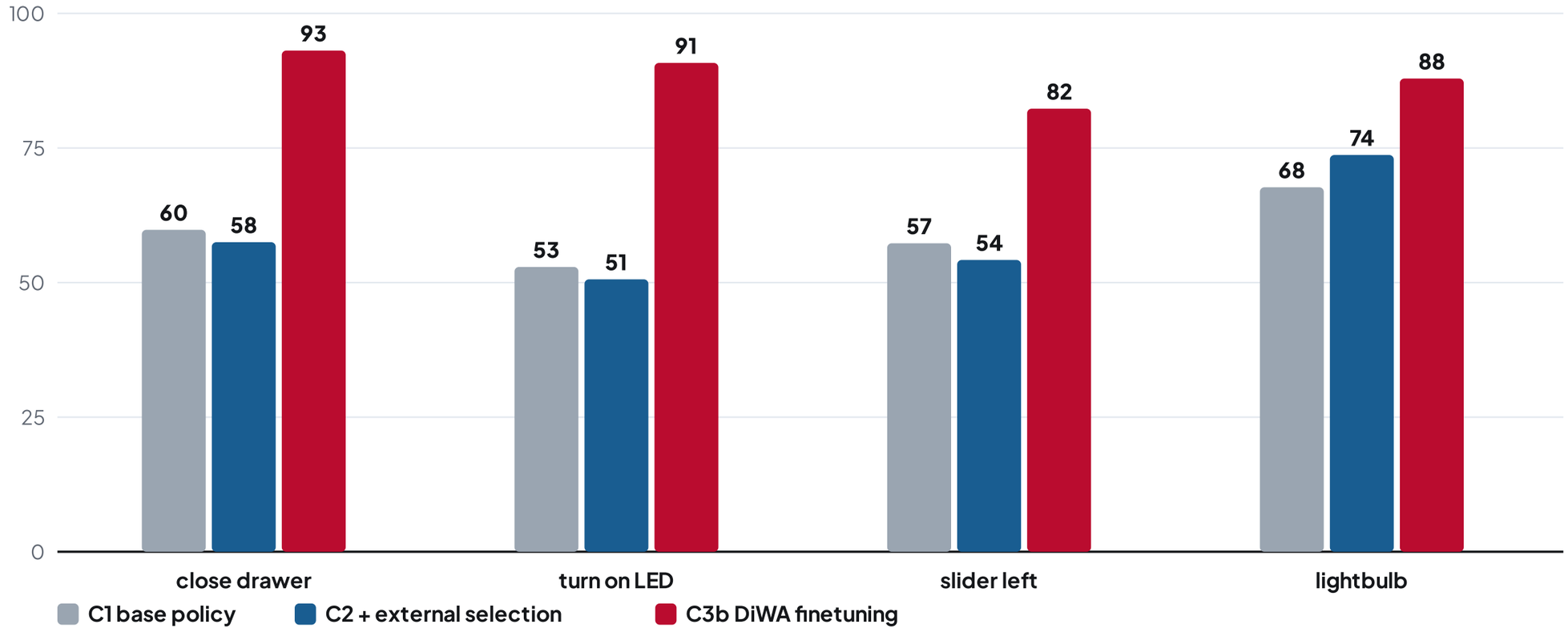
4 tasks
CALVIN MANIPULATION SKILLS

~100 episodes
PER TASK, ABOUT 33 PER SEED

3 seeds
42, 43 AND 44

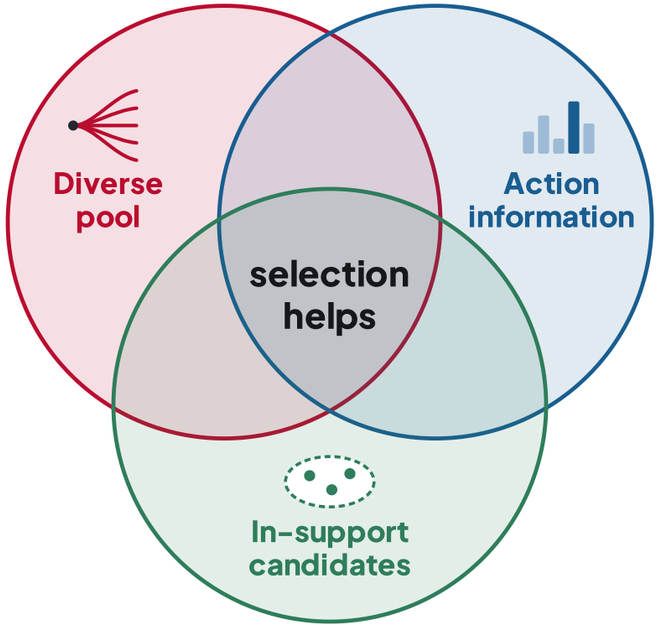
Success rate
% OF EPISODES COMPLETED

Finetuning > external selection



Task success rate (%), CALVIN. Selection (C2) is within noise of the policy alone (C1) on every task; finetuning (C3b) adds 14 to 40 points.

What a world model needs to help



Diverse pool

The candidates must be genuinely different motions, or there is nothing to choose between.

Action information

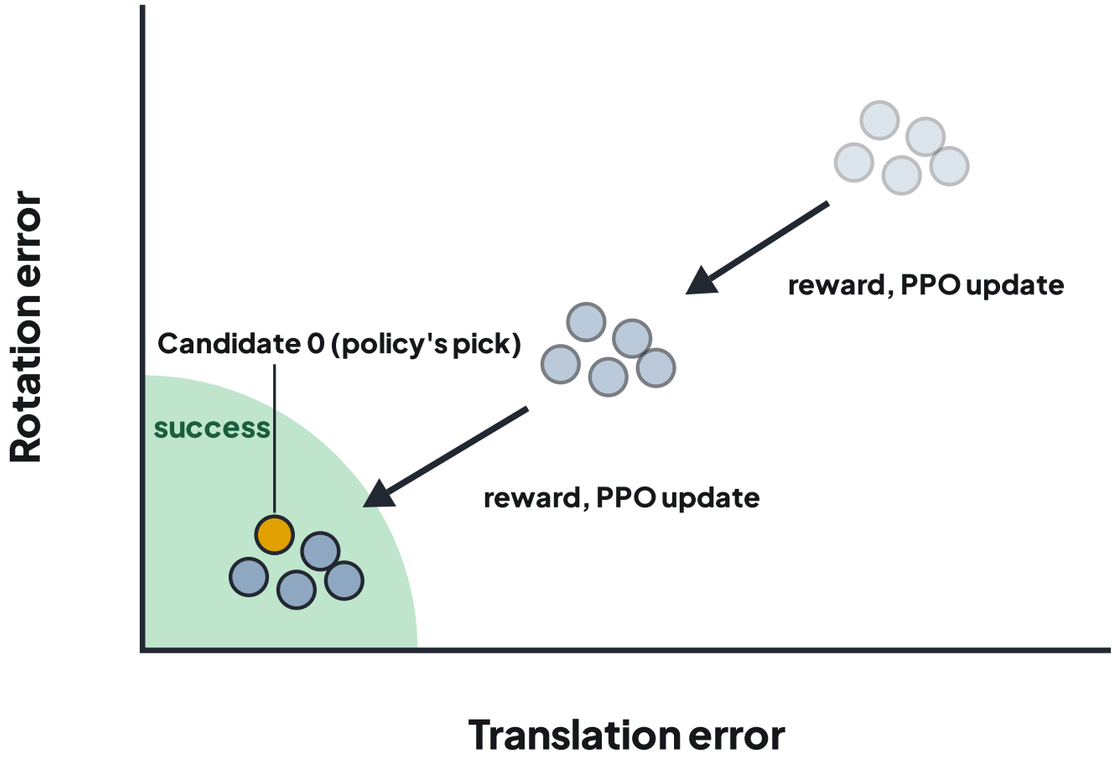
The score must depend on each candidate's motion. Without it, every score ties.

In-support candidates

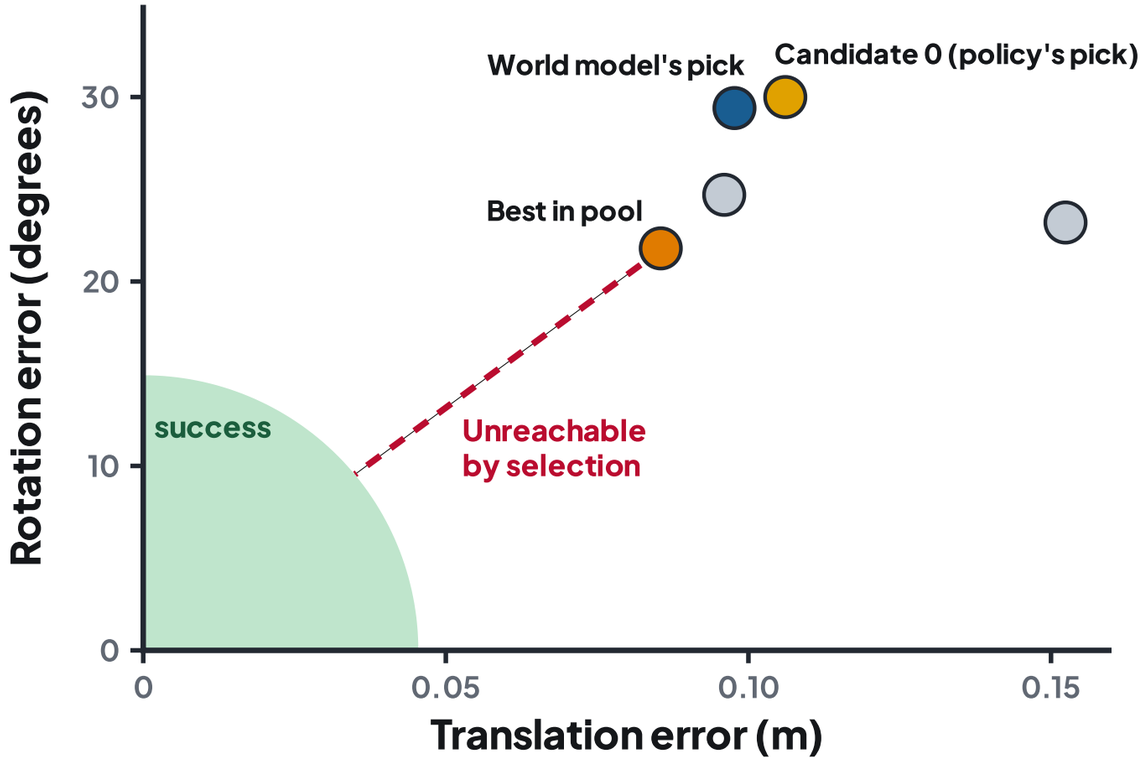
Candidates must be close to the scorer's training data, so its predictions can be trusted.

Diffusion Policy + V-JEPA2-AC ● ● ●
DiWA, by design ● ● ●
ACT, collapsed ○ ● ●
No-action control ● ○ ●
7-D adapters ● ● ○

DIWA (ILLUSTRATIVE)



OUR SELECTION



Finetuning improves the policy. Selection improves the choice.

Conclusions

1. **Yes, for one tested pairing**
   a frozen world model, used externally, selected better motions from the policy's own candidates

2. **Only when all three conditions hold**
   5.98% lower error, about 36% of the headroom, no retraining

3. **Finetuning moves the pool, selection picks from it**
   different tools, not rivals

Limitations

1. **Time and compute**
   K = 5, three seeds, part of EgoDex

2. **Offline and open loop**
   no robot or M.AI.ESTRO deployment yet, so compounding error is untested

3. **Realistic, not successful**
   the scorer judges whether a motion looks realistic; we measured error against recorded human motion, never task success

Future work

1. **Close the loop**
   on a robot and with human guidance, in M.AI.ESTRO

2. **Combine finetuning and selection**
   with separate world-model judges, to get the best of both

3. **Reach the ceiling**
   more data and stronger world models, to capture more of the 16.5% headroom

THANK YOU FOR LISTENING

Questions

Backup slides follow, covering the AI-use workflow, how each method was chosen, LeWM's gate, seeds, pool size, and where selection fails.

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How AI was used in this work

NotebookLM

Queried paper sets for related work and citations, then each source was skimmed manually.

Claude skills

Query, draft and polish workflows, with my own review and edits.

Gemini

Paper discovery and critical review of chapters against a rubric.

Coding agents

Wrote most of the code under my specification, then tested and audited.

WHERE THE DECISIONS SAT

Research question, benchmark design, which methods to adopt, what counted as a valid result, and every claim in the dissertation.

HOW CORRECTNESS WAS CHECKED

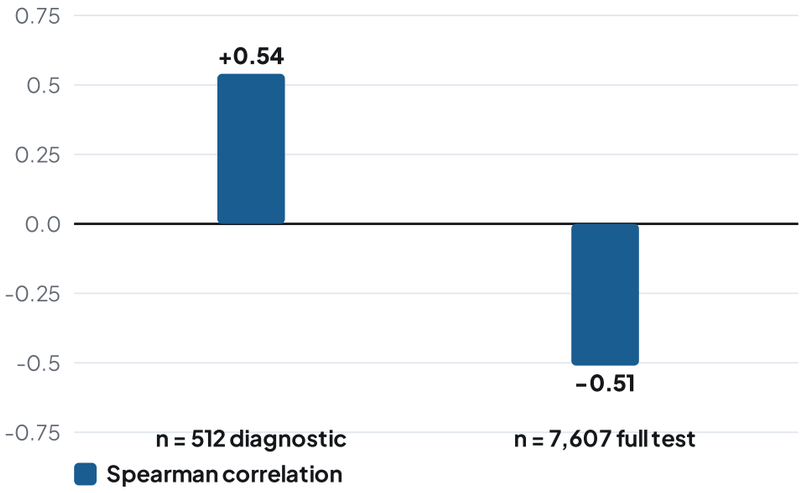
Separate audit passes and re-runs found real defects, including a DDIM sampler indexing error, a training-exposure undershoot, and a DiWA configuration mistake. All were fixed before reporting.

Used most in the literature review and methodology, for example to find established reference methods for handling ACT and LeWM without inventing new verification tests.

Which published work anchored each method choice

CHOICE	ANCHOR	STATUS
Policy-guided candidate search (CEMP)	Cross-entropy method, initialised from the policy output, inspired by TD-MPC2	Adapted, not a published recipe
Posterior-collapse fixes for ACT	KL annealing (Fu et al.), free bits (Kingma et al.), observation dropout	All tried, all failed
Diffusion sampling	DDIM (Song et al.), DP-T architecture (Chi et al.)	Implemented, sampler bug fixed
Action-conditioned scorer	V-JEPA2-AC (Assran et al.)	Head trained locally
Finetuning comparison	DiWA (Chandra et al.) on CALVIN, own checkpoints	Reproduced, secondary study
Pool size K = 5	No literature anchor	Compute choice, ranking stable K = 2 to 20
Matched exposure 100,000 items	Compute ceiling, loss still falling	Not a convergence claim

LeWM passes its ranking gate on 512 windows and fails it on 7,607



WHAT IS AND IS NOT ESTABLISHED

- The gain is real, −10.1% translation, intervals exclude zero
- The reported configuration never invokes LeWM's learned dynamics
- Its score does not predict which candidate is better at full scale
- Reweighting terms to pass the gate was rejected as overfitting the gate

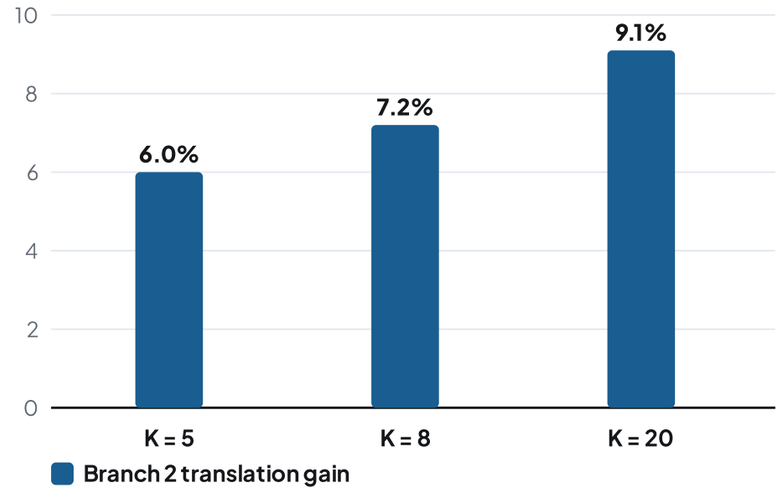
Spearman correlation between score margin and true translation improvement. The 512-window figure was +0.537, the full test set gave −0.51.

The result holds at every seed, and the controls behave as designed

	SEED 42	SEED 43	SEED 44
Baseline translation error	0.1142	0.1178	0.1191
V-JEPA2-AC, Branch 2	-6.41%	-6.09%	-5.46%
V-JEPA2 without action input	0.00%	0.00%	0.00%
PointWorld proxy	+0.50%	+0.23%	+0.07%
LeWM rollout (unvalidated)	-10.10%	-10.06%	-10.12%

Three seeds are a basic robustness check, not proof of stability. DexWM is single-seed by a cost decision of roughly 121 extra GPU-hours.

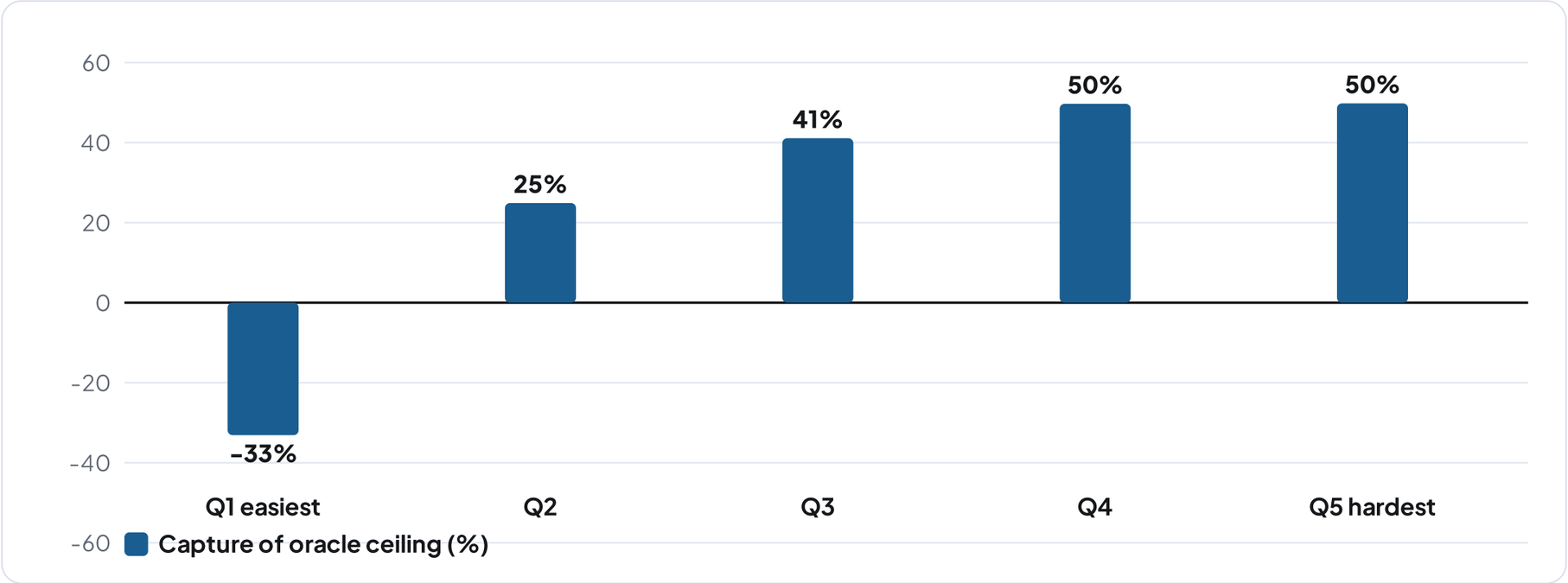
More candidates buy more headroom, not better selection



WHAT THE SWEEP SHOWS

- Scorer ranking identical at every K from 2 to 20
- Gain grows steadily with K and does not plateau
- Capture of the oracle ceiling stays flat, near 40% on these nested pools
- K = 5 was a compute choice, with no literature anchor

On the easiest fifth of windows, selection makes things worse



Windows sorted by baseline error, K = 8, Branch 2. Translation error rises 5.81% in Q1. Confidence gating does not help, because the score gap has a correlation of -0.002 with baseline error. A state-change-targeted windowing removes most of the reversal.

How results were measured

ERROR METRICS

- Translation, root-mean-square per axis, not Euclidean distance
- Rotation, mean geodesic angle in degrees
- Both averaged over the full 16-step chunk
- Change reported relative to candidate 0

STATISTICS

- 95% episode-clustered bootstrap, 2,000 resamples, pooled over seeds
- Oracle ceiling, best candidate in the same pool
- Ranking-validity gate, per-window Spearman correlation
- Win, tie, loss against candidate 0

Test windows are fixed, candidate pools frozen, future-goal access blocked, and scorer support restricted to training data.

What is disclosed about test-set use

- The Branch 2 margin threshold's final value came from a disjoint validation split.
- An earlier, superseded calibration touched the test set directly. This is stated as a named limitation and is reassuring, not a full defence.
- The LeWM gate was tuned on a small diagnostic split and failed to generalise to the full test set. It was reported, not re-tuned.
- Earlier stronger-looking results relying on shortcuts, transductive support or oracle access were corrected rather than reported.

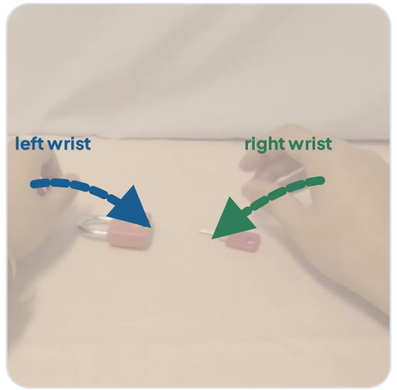
Objective achievement

OBJ.	STATUS	BASIS
1	Fully	Three-branch benchmark implemented as designed and used for every family reported
2	Partly	Local exposure matched, externally pretrained backbones carry an advantage not equalised
3	Within tested configurations	Whether and when selection helps determined for every combination, one positive, a likely explanation for each other

Objectives 4 and 5 are in the Conclusion chapter's objective table. Every non-positive-route explanation is the study's own interpretation, not independently proven.

THE PREDICTION UNIT

A world model predicts what happens next, and a scorer turns that into a choice



Input: the scene and one candidate motion

→ **World model** →



Predicted scene afterwards

→ **Scorer** rates this candidate

The world model does not decide what to do. A prediction is not a decision, so a scorer must turn each predicted future into a score. Illustrative frames: the real prediction is made in a compact latent space, not in pixels.

Our CALVIN reproduction against the DiWA paper

TASK	PAPER BASE	OUR C1	PAPER DIWA	OUR C3B	OUR C2 (SELECTION)
close drawer	59.14	59.77	91.95	93.10	57.47
turn on LED	48.43	52.87	86.21	90.81	50.57
move slider left	62.15	57.29	83.33	82.29	54.17
turn on lightbulb	60.61	67.68	91.92	87.88	73.74

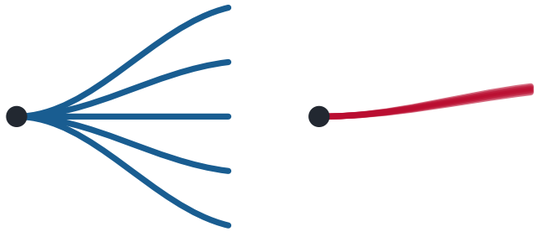
Success rate (%), three seeds. Paper values from Chandra et al. (2025), Table 1. Our runs use the authors' released checkpoints, world model and reward classifier. C1 and C2 share one evaluation harness; C3b is read on DiWA's PPO-trainer loop, so C1 and C3b come from different harnesses. Differences from the paper are within about 7 points and are expected from rerunning on different hardware and seeds. The behaviour-cloning weight α_{BC} was set to the paper's published values after an initial run at zero collapsed two tasks.

The three conditions, case by case

Diffusion Policy with V-JEPA2-AC met all three. Each red case is where one failed.

Diverse pool

✓ five different ✗ five copies



Diffusion Policy ACT, collapsed

The policy must offer genuinely different motions to choose from.

✓ Main pairing: met

Action information

✓ sees the motion ✗ blind to it



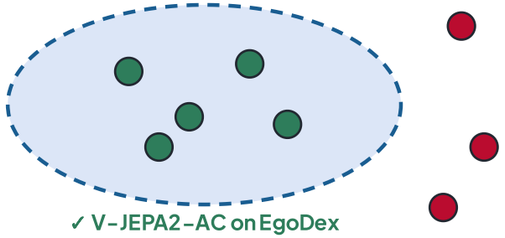
V-JEPA2-AC no-action control: 0.00%

The scorer's prediction must change with each candidate's motion.

✓ Main pairing: met

In-support candidates

what the scorer was trained on ✗ 7-D adapters



✓ V-JEPA2-AC on EgoDex

Candidates must look like the data the scorer learned from.

✓ Main pairing: met

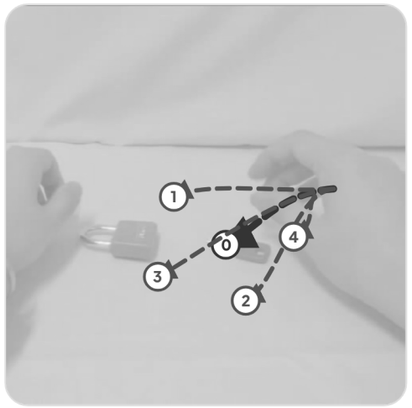
References cited in this deck

- Assran et al. (2025). V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning.
- Chandra et al. (2025). DiWA: Diffusion Policy Adaptation with World Models. CoRL.
- Chi et al. (2023). Diffusion Policy: Visuomotor Policy Learning via Action Diffusion. RSS.
- Fu et al. (2019). Cyclical Annealing Schedule: A Simple Approach to Mitigating KL Vanishing. NAACL.
- Goswami et al. (2025). World Models for Learning Dexterous Hand-Object Interactions from Human Videos.
- Hansen et al. (2024). TD-MPC2: Scalable, Robust World Models for Continuous Control. ICLR.
- Hoque et al. (2026). EgoDex: Learning Dexterous Manipulation from Large-Scale Egocentric Video. ICLR.
- Huang et al. (2026). PointWorld: Scaling 3D World Models for In-The-Wild Robotic Manipulation. CVPR.
- Kingma and Welling (2014). Auto-Encoding Variational Bayes. ICLR.
- LeCun (2022). A Path Towards Autonomous Machine Intelligence, v0.9.2.
- Maes et al. (2026). LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels.
- Mees et al. (2022). CALVIN: A Benchmark for Language-Conditioned Policy Learning for Long-Horizon Robot Manipulation. RA-L.
- Rubinstein and Kroese (2004). The Cross-Entropy Method.
- Song et al. (2021). Denoising Diffusion Implicit Models. ICLR.
- Terver et al. (2026). What Drives Success in Physical Planning with Joint-Embedding Predictive World Models?
- Welch Labs (2026). Yann LeCun's $1B Bet Against LLMs. Video essay.
- Zhao et al. (2023). Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware (ACT). RSS.
- Zhou et al. (2025). DINO-WM: World Models on Pre-Trained Visual Features Enable Zero-Shot Planning. ICML.

DRAFT, not presented

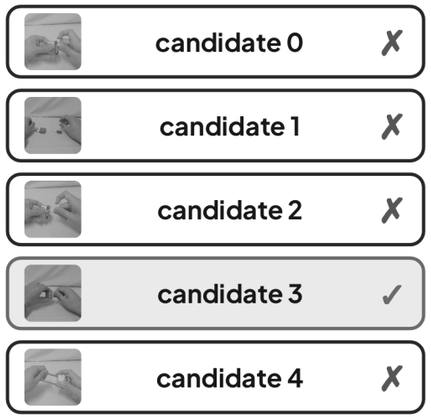
IMAGINE EACH OUTCOME, THEN CHOOSE THE BEST CANDIDATE

How our selection works



Five candidate motions

World model



Imagined outcome of each candidate

$$\arg\max_k s(\hat{o}_k)$$

picks candidate 3

compared with



The real outcome (expert motion)

The real outcome is used only to measure the error. The scorer never sees it.